

The-Anh Vu-Le

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Education

University of Illinois Urbana-Champaign

Urbana, IL

Ph.D. in Computer Science (Advisor: Professor Tandy Warnow)

Aug 2022 – May 2027 (expected)

Research Focus: Network Science, Community Detection, Graph Algorithms

Master in Computer Science

May 2025

Selected Coursework: Deep Learning Theory, Statistical Reinforcement Learning, Deep Generative and Dynamical Models, Network Science, Applied Parallel Programming, Algorithmic Game Theory, Algorithmic Genomic Biology, Formal Methods in Software Development, Abstract Algebra, Real Analysis, Probability Theory.

University of Science, VNU-HCM

Vietnam

B.S. in Computer Science (Valedictorian – top 1/500, Honors Program)

May 2020

Relevant Coursework: Statistical Learning, Big Data, Advanced Probability and Statistics, Computer Vision, Natural Language Processing, Information Retrieval

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, CUDA, HTML/CSS, Julia, Lean

Frameworks: React, Angular, PyTorch, Optuna, D3.js, NumPy, Pandas, Matplotlib

Tools & Infra: Git, Docker, Linux/Bash, CMake, Slurm (HPC), Gurobi, SQLite, $\text{\LaTeX}$

Work Experience

Google

Sunnyvale, CA

Software Engineering Intern

May 2026 – Aug 2026

- Architected a full-stack network telemetry dashboard for the Firefly distributed clock synchronization system, accelerating root-cause analysis for datacenter-scale synchronization anomalies.
- Engineered a high-performance **C++** backend service, implementing optimized query processing pipelines to retrieve and process large-scale topological and telemetry data.
- Built an interactive frontend using **Angular** and **TypeScript**, capable of dynamically rendering large graphs to visualize complex components' health and relationships.

Research Experience

University of Illinois Urbana-Champaign

Urbana, IL

Graduate Research Assistant

Aug 2022 – Present

Accurate and Scalable Graph Analysis | *Advised by Prof. Tandy Warnow*

Jan 2024 – Present

- **Graph generation:** Engineered scalable **Python/C++** software to generate massive benchmark networks with high edge-connectivity community structures (e.g., Edge-Connected Stochastic Block Model) for evaluating detection algorithms [EC-SBM] [RECCS].
- **Community detection:** Designed topology-based partitioning techniques, such as Dense Subgraph Clustering (DSC), which iteratively extracts densest subgraphs to accurately partition vertex sets [DSC].
- **Post-processing:** Developed the Well-Connected Cluster (WCC) technique to refine detection results, specifically mitigating connectivity issues inherent to Stochastic Block Model (SBM) partitions [WCC].
- **Cluster ensemble:** Engineered the Constrained Voting Consensus (CVC) pipeline to aggregate results from multiple

detection techniques by modifying the input network to compute a robust consensus clustering [CVC].

Causal Inference | *Advised by Prof. Hari Sundaram*

Jan 2024 – May 2024

- Evaluated and optimized **stochastic variational inference** algorithms to improve the scalability and accuracy of causal inference modeling pipelines [Report].

Probabilistic Machine Learning | *Advised by Prof. Arindam Banerjee*

Jan 2023 – Dec 2023

- **Stochastic Localization**: Built robust sampling and training pipelines for energy-based models using **PyTorch**. [Code]
- **SGLD Optimization**: Developed a custom mathematical optimization framework for noisy gradient descent leveraging stochastic gradient Langevin dynamics.

Trustworthy Federated Learning | *Advised by Prof. Bo Li*

Aug 2022 – May 2023

- Co-developed **UniFed**, an open-source benchmarking framework for Federated Learning systems, by extending FedML with custom system-testing protocols. [Code]

VinAI Research

Vietnam

Research Resident

Jan 2022 – Aug 2022

Active Learning | *Advised by Prof. Tung Pham*

Apr – Aug 2022

- Engineered an open-source benchmark toolkit in **PyTorch**, establishing a unified evaluation API for active learning algorithms. [Code]

Computational Optimal Transport | *Advised by Prof. Nhat Ho*

Jan – Mar 2022

- Implemented a highly optimized **PyTorch** library for Computational Optimal Transport techniques to accelerate algorithmic research. [Code]

Research Intern

Aug 2021 – Dec 2021

Multiple Object Tracking | *Advised by Prof. Minh Hoai Nguyen*

- Authored and published **trackun**, a highly modular **Python** software package for real-time multiple object tracking utilizing Bayesian filters. [Code]

SELab, University of Science, VNU-HCM

Vietnam

Research Assistant

Jan 2019 – Jul 2021

Computer Vision | *Advised by Prof. Minh-Triet Tran*

- Engineered **RingViewNet**, a novel neural 3D-model embedding system [Code].
- Built scalable codebases for **semantic segmentation** [Code] and **visual re-identification** pipelines [Code].

Publications

* indicates equal contribution

2026: T.-A. Vu-Le, M. Park, I. Chen, T. Warnow. "Using stochastic block models for community detection." *Applied Network Science*. [Paper] [Code]

2026: T.-A. Vu-Le, J.A.C. Lamy, T. Alessi, I. Chen, M. Park, E. Harb, G. Chacko, T. Warnow. "Improving community detection with CVC, a new cluster ensemble method." *Applied Network Science*. [Paper] [Code]

2025: T.-A. Vu-Le, J.A.C. Lamy, T. Alessi, I. Chen, M. Park, E. Harb, G. Chacko, T. Warnow. "Dense Subgraph Clustering and a New Cluster Ensemble Method." *International Conference on Complex Networks and their Applications*. [Paper] [Code]

2025: M. Park, J.A.C. Lamy, E.C.C. Rodrigues, F.M. Ferreira, **T.-A. Vu-Le**, T. Warnow, G. Chacko. "Very Large Scale Simulations of Network Growth with the Scalable Agent-Based Simulator for Citation Analysis with Sampling (SASCA-s)." *International Conference on Complex Networks and their Applications*. [Paper] [Code]

2025: T.-A. Vu-Le, L. Anne, G. Chacko, T. Warnow. "EC-SBM Synthetic Network Generator." *Applied Network Science*. [Paper] [Code]

2025: L. Anne, **T.-A. Vu-Le**, M. Park, T. Warnow, G. Chacko. "RECCS: Realistic Cluster Connectivity Simulator for

Synthetic Network Generation.” *Advances in Complex Systems*. [Paper] [Code]

2024: M. Park, D.W. Feng, S. Digra, **T.-A. Vu-Le**, G. Chacko, T. Warnow. “Improved Community Detection using Stochastic Block Models.” *International Conference on Complex Networks and Their Applications*. [Paper]

2024: L. Anne, **T.-A. Vu-Le**, M. Park, T. Warnow, G. Chacko. “Synthetic networks that preserve edge connectivity.” *International Conference on Complex Networks and Their Applications*. [Paper]

2023: **T.-A. Vu-Le***, C.D. Nguyen*, T. Nguyen, T. Quan, A.T. Luu. “Expand BERT Representation with Visual Information via Grounded Language Learning with Multimodal Partial Alignment.” *ACM International Conference on Multimedia (ACMMM)*. [Paper]

2022: K. Nguyen*, D. Nguyen*, **T.-A. Vu-Le**, T. Pham, N. Ho. “Improving Mini-batch Optimal Transport via Partial Transportation.” *International Conference on Machine Learning (ICML)*. [Paper] [Code]

2021: I. Sipiran, P. Lazo, C. Lopez, . . . , **T.-A. Vu-Le**, et al. “SHREC 2021: Retrieval of cultural heritage objects.” *Computers & Graphics*. [Paper] [Code]

2020: M.-T. Tran, T.V. Nguyen, T.-H. Hoang, . . . , **T.-A. Vu-Le**, et al. “iTASK-Intelligent traffic analysis software kit.” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. [Paper] [Code]

2020: W. Li, D. Song, A. Liu, . . . , **T.-A. Vu-Le**, et al. “SHREC 2020 Track: Extended Monocular Image Based 3D Model Retrieval.” *3DOR: Eurographics Workshop on 3D Object Retrieval*. [Paper]

2019: W. Li, A. Liu, W. Nie, . . . , **T.-A. Vu-Le**, et al. “SHREC 2019-monocular image based 3D model retrieval.” *3DOR: Eurographics Workshop on 3D Object Retrieval*. [Paper]

Talks

2025: “Dense Subgraph Clustering and a New Cluster Ensemble Method.” *International Conference on Complex Networks and Their Applications*. Binghamton University, NY. December 11, 2025.

Personal Projects

2026: Densest Community Search Solver: Architected a full-stack graph analytics platform featuring a high-performance **C++** branch-and-price exact solver (via Gurobi and Dinkelbach updates) for directed citation networks. Engineered a **Python/FastAPI** backend and an interactive **React/D3.js** web UI with live OpenAlex API integration. [Code]

2026: Chess Monte Carlo Simulator: Architected a high-performance **C++** generative modeling and simulation engine for tournament outcome prediction, sampling millions of possible futures via dynamic Bayesian rating updates and a parametric draw model. Integrated with a **Python/Optuna** hyperparameter tuning pipeline and an interactive **JavaScript/HTML** analytics dashboard. [Code]

2025: Neural Phylogenetic Tree Estimation: Engineered a **PyTorch**-based metric learning pipeline for efficient large-scale DNA sequence processing. [Code]

2025: CUDA LeNet Optimizer: Accelerated forward convolution kernels using advanced GPU memory tiling and thread-level parallelization in **CUDA/C++**. [Code]

2025: Lean Math Formalization: Translated complex mathematical textbooks into verified programmatic proofs using **Lean 4** and mathlib4. [Code]

2022: PRNG Library: Implemented a robust algorithmic collection of pseudo-random number generators and statistical sampling techniques in **Python**. [Code]

2020: Battleship AI Engine: Built a **C#/Unity** simulation and visualization environment for training and evaluating adversarial AI agents. [Code]

2019: Torchann Framework: Architected a modular, extensible training and evaluation software framework wrapping **PyTorch** for deep neural networks. [Code]